

UNIT-3: Ordinary differential equations of higher orders

PART – A (5*2=10)

A1: [CO3] Solve: $(D^2 + 6)y = \sin 4x$

A2: [CO3] Write the general formula P.I. of $f(D)y = x.V$ where V is function of x.

A3: [CO3] Write the general formula P.I. of $f(D)y = e^{ax}.V$ where V is function of x.

A4: [CO3] Define homogeneous differential equation of second order.

A5: [CO3] Define variation of parameter method.

PART – B (3*7=21)

B1: [CO3] Solve: $(D^2 - 1)y = x \sin 3x + \cos x$.

$$\text{Ans. } y = C_1 e^x + C_2 e^{-x} - \frac{x}{10} \sin(3x) - \frac{3}{50} \cos(3x) - \frac{1}{2} \cos(x)$$

B2: [CO3] Solve: $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = 2 \log x$.

$$\text{Ans. } y = x(C_1 + C_2 \log x) + 2(2 + \log x)$$

B3: [CO3] Solve by variation of parameter method: $\frac{d^2 y}{dx^2} - y = \frac{2}{1+e^x}$.

$$\text{Ans. } y = C_1 e^x + C_2 e^{-x} + e^x \log \left(\frac{1+e^x}{e^x} \right) - 1 - e^{-x} \log(1 + e^x)$$

PART – C (3*11=33)

C1: [CO3] Solve $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = e^x$

$$\text{Ans. } y = c_1 x^{-1} + c_2 x^{-2} + x^{-2} e^x$$

C2: [CO3] Solve: $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 2x^2$.

$$\text{Ans. } y = (C_1 + C_2 \log x)x^2 + (x + \log x)^2$$

C2: [CO3] Solve by variation of parameter method: $\frac{d^2 y}{dx^2} + a^2 y = \sec ax$.

$$\text{Ans. } y = C_1 \cos ax + C_2 \sin ax + \frac{x}{a} \sin ax + \frac{\log \cos ax}{a^2} \cos ax$$
